Physics P3: Solutions

Gravity, Orbits, and Waves · MAIS PAC

Problem 1.

(a) A satellite's orbital plane stays fixed relative to the distant stars (inertial space), so to hover over a fixed point on Earth's surface it must match Earth's true 360° rotation period relative to the stars—one sidereal day (86,164 s). Moreover, Earth's gravity pulls toward Earth's center, so every Keplerian orbit lies in a plane passing through Earth's center (a great circle). Hovering perpetually at latitude 47.9° N would define a small circle north of Earth's center whose centripetal acceleration points toward Earth's rotation axis rather than Earth's center, leaving an unbalanced southward component of gravity.

(b) From Kepler's third law $\mathbf{r} = (\mathbf{GM}_{\oplus}\mathbf{P}^2 / 4\pi^2)^{1/3}$, using $\mathbf{GM}_{\oplus} = 3.986 \times 10^{14}$ m^3/s^2 and $\mathbf{P} = 86{,}164$ s:

$$\mathbf{r} = (((3.986 \times 10^{14})(86{,}164)^2)/ (4\pi^2))^{1/3} \approx 4.216 \times 10^7 \text{ m} = 42{,}160 \text{ km}. \quad (1)$$

Subtracting $\mathbf{R}_{\oplus} = 6{,}370$ km gives an altitude $\mathbf{h} \approx$ **35,790 km** (about 5.6 Earth radii above the equator).

Problem 2.

(a) For a circular orbit of radius $\mathbf{r}$, kinetic energy is $\mathbf{K} = +\mathbf{GMm}/(2\mathbf{r})$, potential energy is $\mathbf{U} = -\mathbf{GMm}/\mathbf{r}$, and total energy is $\mathbf{E} = -\mathbf{GMm}/(2\mathbf{r}) = -\mathbf{K}$. When drag does negative work $\mathbf{W}_{\text{drag}} = \Delta\mathbf{E} < 0$, total energy $\mathbf{E}$ becomes more negative, so $\mathbf{r}$ decreases—and therefore kinetic energy $\mathbf{K} = -\mathbf{E}$ **increases** by $\Delta\mathbf{K} = -\Delta\mathbf{E} > 0$! The satellite **speeds up** as it spirals lower.

(b) Because the satellite spirals slightly inward rather than moving strictly perpendicular to gravity, its velocity vector has a tiny downward radial component toward Earth's center. Gravity therefore does **positive** work $\Delta\mathbf{W}_{\text{grav}} = -\Delta\mathbf{U} = +2|\Delta\mathbf{E}|$, twice as large as the energy $-|\Delta\mathbf{E}|$ dissipated by drag, leaving net kinetic energy gain $\Delta\mathbf{K} = +|\Delta\mathbf{E}|$.

Problem 3.

(a) Since $\mathbf{r_p} + \mathbf{r_a} = 2\mathbf{a}$, the semi-major axis is $\mathbf{a} = (1.00 + 1.52)/ (2) =$ **1.26 AU**. A full orbit on this ellipse takes $\mathbf{P} = \mathbf{a}^{3/2} = (1.26)^{3/2} \approx 1.414$ yr. Coasting from perihelion (Earth) to aphelion (Mars) is exactly half an orbit:

$$\mathbf{t}_{\text{transfer}} = (\mathbf{P})/ (2) \approx 0.707 \text{ yr} \times 365.25 \text{ d/yr} \approx 258 \text{ days} (\approx 8.5 \text{ months}). \quad (2)$$

(b) At $\mathbf{r} = \mathbf{r}_1$, Earth's circular speed satisfies $\mathbf{v}_{\oplus}^{\,2} = \mathbf{GM}_{\odot}/\mathbf{r}_1$. Dividing the vis-viva equation by $\mathbf{v}_{\oplus}^{\,2}$ gives

$$(\mathbf{v_p})/ (\mathbf{v}_{\oplus}) = \sqrt{(2 - (\mathbf{r}_1)/ (\mathbf{a}))} = \sqrt{(2 - (1.00)/ (1.26))} \approx 1.098.$$

(3)

Thus $\mathbf{v_p} \approx 1.098 \times 29.8$ km/s ≈ 32.7 km/s, requiring a departure boost $\Delta\mathbf{v}_1 = \mathbf{v_p} - \mathbf{v}_\oplus \approx$ **2.9 km/s**.

Problem 4.

(a) Around the center of mass, momentum vanishes: $\mathbf{M}_\star\mathbf{v}_\star = \mathbf{m_p}\mathbf{v_p}$. For $\mathbf{m_p} \ll \mathbf{M}_\star$, the planet's orbital radius $\mathbf{a}$ satisfies Kepler's third law $\mathbf{a} = (\mathbf{GM}_\star\mathbf{P}^2 / 4\pi^2)^{1/3}$, so its circular speed is $\mathbf{v_p} = 2\pi\mathbf{a}/\mathbf{P} = (2\pi\mathbf{GM}_\star / \mathbf{P})^{1/3}$. Multiplying by $\mathbf{m_p}/\mathbf{M}_\star$ immediately gives

$$\mathbf{v}_\star = (\mathbf{m_p})/(\mathbf{M}_\star)\,((2\pi\mathbf{GM}_\star)/(\mathbf{P}))^{1/3} = \mathbf{m_p}\,((2\pi\mathbf{G})/(\mathbf{M}_\star{}^2\mathbf{P}))^{1/3}. \tag{4}$$

(b) With $\mathbf{P} = 11.86 \times 3.156 \times 10^7$ s $\approx 3.74 \times 10^8$ s, Jupiter's orbital speed is $\mathbf{v_J} = (2\pi\mathbf{GM}_\odot/\mathbf{P})^{1/3} \approx 1.306 \times 10^4$ m/s. Multiplying by $\mathbf{m_J}/\mathbf{M}_\odot = 9.55 \times 10^{-4}$ yields $\mathbf{v}_\odot \approx$ **12.5 m/s** (a brisk bicycle ride!), producing a tiny Doppler shift $\Delta\lambda/\lambda_0 = \mathbf{v}_\odot/\mathbf{c} \approx$ **4.2×10^{-8}**. Modern stabilized spectrographs measure shifts down to ~0.3 m/s, routinely discovering rocky planets around nearby stars.